AMRUTA INSTITUTE OF ENGINEERING AND MANAGEMENT SCIENCES – POLYTECHNIC

Course: Database System Concepts and PL/SQL

Course Code: 20CS34P

Week 13

Managing and Controlling Transactions

➤ 1. Introduction to Transaction

- A **Transaction** is a sequence of one or more SQL operations (like Insert, Update, Delete, Select) executed as a single unit of work.
- It ensures data consistency in a database system.
- Example: Transferring money from one bank account to another involves multiple steps (debit one account, credit another), and all must succeed or fail together.

- **2. ACID Properties of Transaction**
- ACID ensures reliability in transaction processing.

Property	Description
A – Atomicity	All operations of a transaction succeed or none are applied. (All or nothing)
C – Consistency	Transaction must take the database from one valid state to another, maintaining data integrity.
I – Isolation	Transactions execute independently without interference. Intermediate states are not visible to others.
D – Durability	Once committed, the changes are permanent, even in case of system failures.

3. States of a Transaction

State	Description
Active	Transaction starts and is executing SQL statements.

State	Description
Partially Committed	Last statement executed, but not yet fully committed.
Committed	All changes are saved permanently in the database.
Failed	An error occurs, and the transaction cannot proceed.
Aborted	The transaction is rolled back to ensure data consistency.

4. Transaction Control Commands

1 COMMIT

- Saves all changes made by the transaction to the database permanently.
- Example:

```
INSERT INTO accounts (account_no, balance) VALUES (101, 5000);
COMMIT;
```

2 ROLLBACK

- Undoes all changes made by the transaction since the last COMMIT.
- Example:

```
UPDATE accounts SET balance = balance - 1000 WHERE account_no = 101;
ROLLBACK;
```

3 SAVEPOINT

- Sets a point within a transaction to which we can later roll back, without rolling back the entire transaction.
- Example:

```
SAVEPOINT sp1;
UPDATE accounts SET balance = balance - 500 WHERE account_no = 101;
```

```
-- Now rollback to savepoint
ROLLBACK TO sp1;
```

4️⃣ SET TRANSACTION

- Defines characteristics of the transaction (e.g., isolation level).
- Example:

```
SET TRANSACTION ISOLATION LEVEL SERIALIZABLE;
```

5. Overview of Transaction Management

- Ensures data consistency, even in cases of system failures.
- Transaction Manager handles all transactions ensuring **ACID properties**.
- Typical Workflow:
  1. Begin Transaction
  2. Execute SQL Commands
  3. Savepoint (optional)
  4. Commit / Rollback as needed

6. Example Scenario

Example: Bank Transfer

```
BEGIN TRANSACTION;
UPDATE accounts SET balance = balance - 1000 WHERE account_no = 101; -- Debit
UPDATE accounts SET balance = balance + 1000 WHERE account_no = 102; -- Credit
-- Commit the transaction if both updates succeed
COMMIT;
-- Or Rollback if any update fails
ROLLBACK;
```

With SAVEPOINT:

```
BEGIN TRANSACTION;

UPDATE accounts SET balance = balance - 1000 WHERE account_no = 101;

SAVEPOINT sp1;

UPDATE accounts SET balance = balance + 1000 WHERE account_no = 102;

-- Suppose second update fails

ROLLBACK TO sp1;  -- Undo only the second update

COMMIT;
```